

MAKKAH, Saudi Arabia

WMO: 410300

Lat: 21.437N

Lon: 39.770E

Elev: 240

StdP: 98.47

Time Zone: 3.00 (E03)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.0	18.2	3.2	4.9	27.3	5.1	5.6	28.9	6.7	27.8	5.5	27.5	1.1	0	0.395

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	45.2	24.5	44.2	24.5	43.2	24.4	29.2	38.7	28.4	38.0	27.7	37.5	2.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	23.4	35.5	26.0	22.0	35.1	25.1	20.8	34.6	97.7	38.3	93.7	38.0	90.5	37.7	34.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.6	4.9	4.1	DB	14.5	48.3	1.6	1.1	13.3	49.1	12.4	49.8	11.5	50.4	10.3	51.2	
			WB	9.8	30.1	1.1	1.2	8.9	31.0	8.3	31.7	7.7	32.3	6.8	33.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.9	25.1	26.5	28.4	31.9	35.2	36.6	36.5	36.5	36.1	33.4	29.5	26.7
	DBStd	4.65	2.09	2.27	2.41	2.17	2.06	1.63	1.39	1.30	1.29	1.69	1.93	1.81
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7992	467	461	571	657	782	799	823	822	783	725	585	517
	CDD18.3	4951	209	228	312	407	524	549	564	564	533	467	335	258
	CDH23.3	72041	1699	2143	3589	5793	8471	9334	9662	9437	8621	6898	3985	2407
Wind	WSAvg	48177	679	993	1912	3629	6030	6937	7183	6957	6221	4478	2092	1066
	PrecAvg	1.2	1.2	1.3	1.4	1.4	1.3	1.2	1.2	1.3	1.1	1.0	1.0	1.0
	PrecMax	106	16	2	5	11	7	1	2	8	6	13	24	12
	PrecMin	230	57	40	32	88	43	3	21	57	39	70	176	53
	PrecStd	26	0	0	0	0	0	0	0	0	0	0	0	0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	34.2	36.2	39.2	42.1	46.8	47.1	46.1	45.2	45.8	43.9	38.8	35.2	
	MCWB	21.6	22.3	22.1	23.6	24.4	23.5	24.2	26.4	24.6	23.1	22.7	21.7	
	DB	33.0	35.0	37.2	40.6	44.2	45.4	44.2	43.8	44.1	42.1	37.2	34.1	
	MCWB	21.5	22.4	22.1	23.4	24.1	24.2	24.4	25.8	24.7	23.4	23.2	22.1	
	DB	32.0	33.9	36.1	39.3	43.0	44.1	43.1	42.8	42.9	40.9	36.2	33.1	
	MCWB	21.3	22.1	22.1	23.4	23.9	24.2	24.1	25.5	25.2	23.7	23.1	22.1	
	DB	30.7	32.2	34.8	38.1	41.7	43.0	42.1	41.8	41.8	39.2	35.1	32.0	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCWB	21.0	21.5	21.8	22.9	23.7	24.1	23.8	25.2	25.4	24.1	23.3	21.9	
	WB	24.1	24.9	25.2	26.5	28.6	28.6	29.2	30.2	30.1	28.1	26.1	25.1	
	MCDB	30.0	33.6	34.2	37.7	40.8	41.0	39.0	40.3	39.8	37.4	33.8	30.9	
	WB	23.2	24.0	24.2	25.5	26.8	27.3	28.0	29.2	29.1	27.0	25.4	24.2	
	MCDB	29.3	31.8	33.4	36.7	38.9	40.3	38.1	38.5	38.5	35.7	32.8	30.1	
	WB	22.7	23.2	23.5	24.8	25.9	26.3	26.9	28.5	28.2	26.3	24.8	23.6	
	MCDB	28.7	30.6	32.6	35.7	38.2	39.8	37.7	37.6	37.7	34.8	32.2	29.6	
Mean Daily Temperature Range	WB	22.0	22.6	22.9	24.2	25.1	25.5	26.0	27.8	27.5	25.8	24.3	23.0	
	MCDB	27.8	29.5	31.7	35.0	37.9	39.2	37.6	36.9	37.0	34.3	31.5	29.2	
	MDBR	10.8	11.7	12.8	13.1	13.5	13.7	12.4	11.7	12.7	13.0	11.4	10.9	
	5% DB	12.4	13.1	13.9	14.0	14.8	14.9	13.4	12.8	14.0	14.5	12.6	12.1	
	MCWBR	5.3	5.3	4.9	5.0	5.1	5.2	5.0	5.0	4.8	4.3	4.0	4.9	
Clear-Sky Solar Irradiance	5% WB	10.7	11.7	12.0	12.4	12.9	13.2	11.3	11.1	11.8	12.6	11.2	10.6	
	MCWBR	4.7	5.1	5.2	5.2	5.4	5.7	5.3	5.0	5.2	4.2	3.9	4.4	
	taub	0.439	0.483	0.509	0.542	0.589	0.608	0.639	0.664	0.618	0.545	0.478	0.449	
	taud	2.129	1.990	1.914	1.862	1.745	1.697	1.673	1.632	1.698	1.850	2.023	2.095	
All-Sky Solar Radiation	Ebn at Noon	820	807	805	783	741	721	699	683	706	740	774	792	
	Edh at Noon	143	174	196	209	233	242	248	259	239	198	158	144	
All-Sky Solar Radiation	RadAvg	4.59	5.49	6.51	7.09	7.52	7.80	7.42	6.96	6.56	5.90	4.88	4.39	
	RadStd	0.24	0.24	0.24	0.27	0.24	0.17	0.18	0.16	0.19	0.17	0.21	0.19	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
Station Only		(d) +0.49	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) +215	(m) +216
Regional (2 neighbors)		+0.48	+0.70	-1.24	N/A	N/A	N/A	N/A	N/A	+188	+172

Nomenclature: See separate page